

Question:

John plans to sell all his toys and use the money to buy video games. He has 13 lego sets and he sells them for \$15 each. He ends up buying 8 video games for \$20 each and has \$5 left. How many lego sets does he still have?

Reasoning + Answer:

<tool_call> Reasoning process here :

Step 1 : Determine the total money John earned from selling the Legos .
He has 13 Lego sets and sells each for \$ 15 .
So , total money earned = 13 × 15

Calculate that :
 $13 \times 15 = (10 \times 15) + (3 \times 15) = 150 + 45 = 195$
So , John earned \$ 195 from selling the Legos .

Step 2 : Determine how much he spent on video games .
He buys 8 video games at \$ 20 each .
Total spent = $8 \times 20 = 160$

Step 3 : He has \$ 5 left after buying the video games .
So , total money spent = amount earned − money left
 $= 195 - 5 = 190$

But this should also be equal to what he spent on video games : \$ 160
Wait − there 's a contradiction here . He spent \$ 160 and has \$ 5 left , so total money spent is $160 + 5 = 165$.

But that doesn 't match his total earnings of \$ 195 . So , perhaps he had \$ 195 , spent \$ 160 on video games , and has \$ 35 left , not \$ 5 .

Wait − let 's go back to the problem :
He sells 13 Lego sets for \$ 15 each − $13 \times 15 = \$ 195$
Buys 8 video games at \$ 20 each − $8 \times 20 = \$ 160$
Has \$ 5 left .

Total money used = $160 + 5 = \$ 165$

But he earned \$ 195 − So he has $195 - 165 = \$ 30$ left? But problem says he has \$ 5 .

So discrepancy suggests a different interpretation .

But the question is : " How many lego sets does he still have ?"

He sells all his toys , so " he still has " implies the number of Legos left after selling ? But if he sells all of them , he has 0 left .

But that seems too literal .

But perhaps the problem has a typo or logical flaw ?

Wait − let 's reread :

" John plans to sell all his toys and use the money to buy video games . He has 13 lego sets and he sells them for \$ 15 each ."

So he sells all 13 − so he sells all of them .

He uses the money to buy video games , and ends up buying 8 video games for \$ 20 each and has \$ 5 left .

So : total money from selling legos = $13 \times 15 = \$ 195$

Money spent on video games = $8 \times 20 = \$ 160$

Money left = $\$ 195 - \$ 160 = \$ 35$

But problem says he has \$ 5 left .

That contradicts .

Therefore , either :

- The number of legos is not 13 , or
- The price per lego is different , or
- The number of video games or price is wrong , or
- The left over money is wrong .

But the problem states :
- 13 lego sets
- \$ 15 each
- 8 video games at \$ 20
- \$ 5 left

So total money from selling = $13 \times 15 = 195$

Money spent = $8 \times 20 = 160$

Money left = $195 - 160 = 35$ − not 5 .

So the numbers are inconsistent .

Therefore , unless he did not sell all of the legos , the contradiction remains .

But the problem says " he plans to sell all his toys " and " he sells them " , meaning all .

But according to the numbers ,